PRACTICAL:2

Aim :

Theory :

Code :

```
clc;
clear;
close all;
a = imread('charlie.jpeg');
figure, imshow(a);
b = rgb2gray(a);
figure, imshow(b);
impixelinfo
%% Image Cropping
c = b(49:159,52:149);
figure, imshow(c);

%% Image Rotation
d = c';
g = d';
figure, imshow(d);
```

```
figure, imshow(g);
%% Display Images in Sequence
e = imresize(d,[111,98]);
f = horzcat(c,e);
figure, imshow(f);
%% Divide Image into RGB Channels
red = a(:,:,1);      % Red channel
green = a(:,:,2);    % Green channel
blue = a(:,:,3);     % Blue channel
cI = cat(3,red,green,blue);
figure;
subplot(1,4,1);
imshow(red);
title('Red Channel');
subplot(1,4,2);
imshow(green);
title('Green Channel');
subplot(1,4,3);
imshow(blue);
title('Blue Channel');
subplot(1,4,4);
imshow(cI);
title('Color Image');
%% Make 4D Data
img4D = cat(4, red, green, blue);
figure;
subplot(1,3,1);
imshow(img4D(:,:,:,1));
title('4D Slice - Red');
subplot(1,3,2);
imshow(img4D(:,:,:,2));
title('4D Slice - Green');
subplot(1,3,3);
imshow(img4D(:,:,:,3));
title('4D Slice - Blue');
disp('Size of 4D Data:');
disp(size(img4D));
```

Output:





Conclusion: